

TerraScout

Robotics software simulation by Kazmir Fahrier

Summary

An orchard robotics simulation by Kazmir Fahrier. Estimated pose drives goal service missions with independent swept collision checks. The project demonstrates software integration; hardware, field accuracy, and production safety remain unvalidated.

Implemented Stack

Control	PID feedback	Measured speed feedback and checked waypoint lookahead
Tracking	Kalman filter	Synthetic worker observations transformed by selected pose
Localization	Particle filter	Default navigation against a known tree map
Mapping	EKF SLAM	Alternate pose source and independent error reporting
Planning	Grid and Hybrid A*	Inflated footprints, checked segments, explicit failures
Scheduling	Exact resource search	Goal priorities and separately enforced execution budgets
Evaluation	Swept contact checks	Trees, independent workers, and orchard boundaries
Sensors	Synthetic observations	Local labeled centroids, encoders, and IMU yaw rate

Evidence and Scope

Default navigation	Particle filter
Goal completion	Includes all requested goals in denominator
Collision evaluation	Full body and swept motion
Resource accounting	Actual distance, elapsed time, modeled energy
Benchmark evidence	README and archived reference summary
Raw scan perception	Separate modules; not integrated in mission
Field validation	Not performed
Authorship	Kazmir Fahrier

Reproduce

```
python -m pip install -e ".[dev]" -c constraints-ci.txt  
  
terrascout-reproduce --skip-gif  
  
python -m pytest
```

Current Roadmap

Integrate raw scan perception and explicit detection failures. Validate calibrated hardware sensors and actuators. Evaluate independent worker encounters and recovery policies.